Case Study 6: Password Policy Validation Engine
Complete Python Source Code

```
passwords = ["Hello@123", "passwordA1!", "abc111XY!", "short"]
prohibited = "password"
for password in passwords:
    failures = []
    if len(password) < 8:
        failures.append("must contain at least 8 characters")
    if not any(ch.isupper() for ch in password):
        failures.append("needs an uppercase letter")
    if not any(ch.islower() for ch in password):
        failures.append("needs a lowercase letter")
    if not any(ch.isdigit() for ch in password):
        failures.append("needs a digit")
    if not any(not ch.isalnum() for ch in password):
        failures.append("needs a special character")
    if prohibited in password.lower():
        failures.append("contains prohibited word 'password'")
    repeated = False
    for i in range(len(password) - 2):
        if password[i] == password[i + 1] == password[i + 2]:
            repeated = True
    if repeated:
        failures.append("has 3 identical consecutive characters")
    print(f"Password: {password}")
    if failures:
        print("Status: FAIL")
        for failure in failures:
            print("-", failure)
    else:
        print("Status: PASS")
    print()
```